

CO2-AT2

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Course

code: CSAI316

Q1 (a)

data mining:

data mining is the process of extracting useful patterns, relationships and knowledge from large datasets.

any two data mining tasks:

1. classification

2. clustering

steps in the KDD process

1.) Data selection

2.) Data cleaning

3.) Data integration

4.) Data transformation

5.) Data mining

6.) Pattern Evaluation

7.) Knowledge Presentation

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b).

Data cleaning:
Data cleaning is the process of detecting and correcting inaccurate, incomplete, or inconsistent data.

Q2(a)

Data Integration:

Combining data from multiple sources into a single unified dataset.

Challenges:

- 1) Data redundancy
- 2) Data inconsistency

b) Covariance Analysis:

Two covariance

- 1) Positive covariance
- 2) Negative covariance

Q3(a)

Data: A = 2, 3, 5, 4, 6, B = 5, 8, 10, 11, 12

Both values generally increase together.

- * Covariance is positive.
- * stocks rise together.
- * They are not independent.

Q3(b)

- * Data Sources $\rightarrow$ store raw data.
- * Database / Data warehouse - central repositories
- * Data cleaning - Removes errors.
- * Data Mining Engine $\rightarrow$ Discover Patterns.
- * Pattern Evaluation $\rightarrow$ Finds useful patterns.

Q4(a)

$$X = \{4, 5, 6, 7, 9\}$$

$$Y = \{8, 3, 7, 5, 6\}$$

$$\text{mean}(x) = 6.2$$

$$\text{mean}(y) = 5.8$$

$$\text{Covariance} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1} = \frac{-4.8}{4} = -1.2$$

Answer: Covariance = -1.2

Q5.)

dataset:

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 30,
33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70

using Equal-Frequencies (Equal-Depth)

Binning (3 bins):

bin 1: 13, 15, 16, 16, 19, 20, 20, 21, 22

smoothed $\rightarrow$ 13, 13, 13, 13, 22, 22, 22, 22, 22

bin 2: 22, 25, 25, 25, 25, 30, 33, 33, 35

smoothed $\rightarrow$ 22, 22, 22, 22, 22, 35, 35, 35

Bin 3: 35, 35, 36, 40, 45, 46, 52, 70, 35

sorted: 35, 35, 35, 36, 40, 45, 46, 52, 70

smoothed $\rightarrow$ 35, 35, 35, 35, 35, 70, 70, 70